Project Design Phase-II

Technology stack (Architecture & Stack)

Date	8 April 2026
Team ID	SWTID-2026-7471
Project name	Automating employee data
Maximum marks	4 marks

Technology stack

A **technology stack** is the combination of tools, frameworks, programming languages, and architecture used to build and run an application.

Common Types of Architecture

1. Monolithic Architecture

- Entire application is built as a single unit
- Simple but hard to scale

Example: Small college project systems

2. Client–Server Architecture

- Frontend (client) communicates with backend (server)

Example: Web applications

3. 3-Tier Architecture

- **Presentation Layer** → UI (Frontend)
- **Application Layer** → Business Logic (Backend)

4. Microservices Architecture

- Application split into small independent services
- Each service handles a specific function

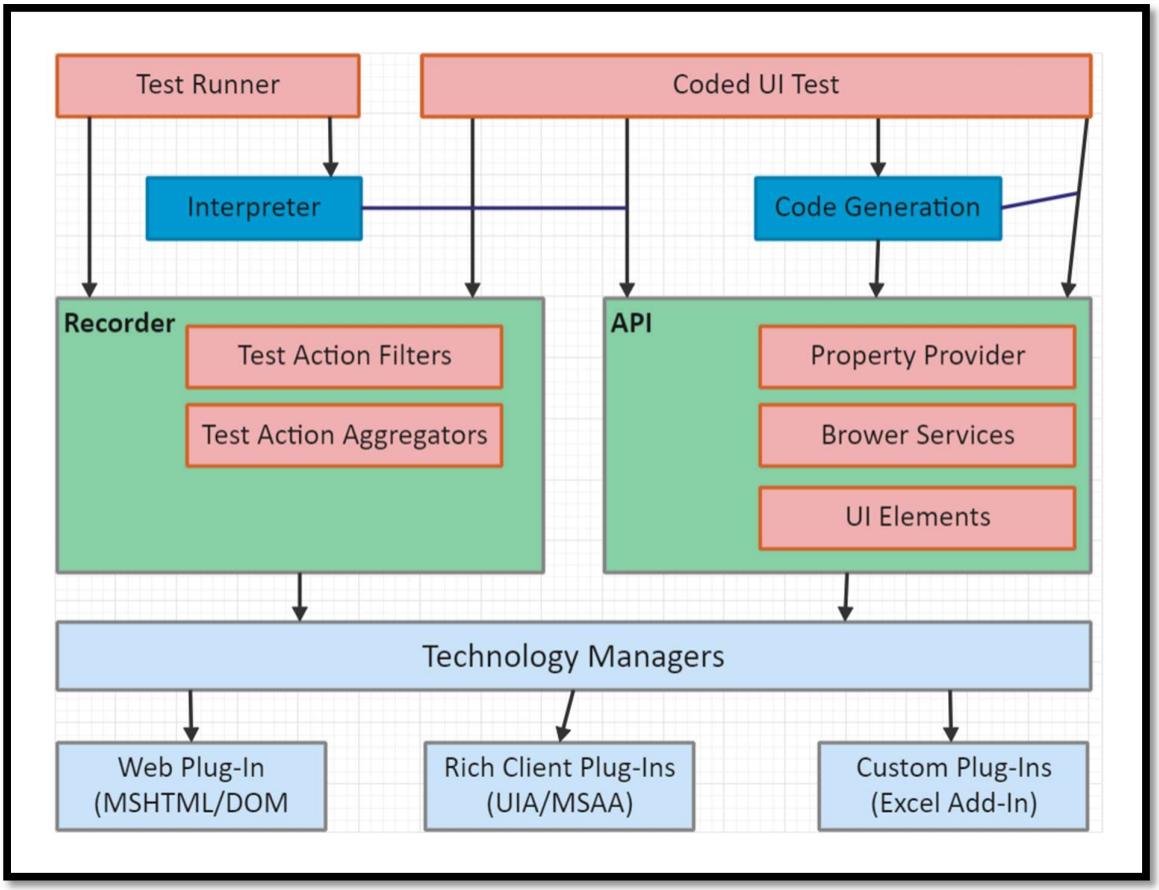
Example: Large-scale apps like e-commerce or streaming platforms

5. Cloud-based Architecture

- Hosted on cloud platforms
- Scalable and flexible

Key Features Supported by This Stack

- Employee record management
- Attendance tracking
- Payroll automation
- Leave management
- Role-based dashboards



Conclusion :

Overall, this technology stack enables seamless automation of employee-related processes like record management, attendance, and payroll, reducing manual effort and errors while improving organizational productivity.